Yongjae Lee

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Biography

Yongjae Lee is an Associate Research Engineer at the Industrial Artificial Intelligence Research Institute at Pusan National University, Republic of Korea. His research interests include business process management and data science, with particular focus on designing tools and architectures for process intelligence. He received his bachelor's degree in Industrial Engineering from Pusan National University in 2024 and received his master's degree in Industrial Data Science & Engineering from Pusan National University in 2026 under Prof. Hyerim Bae's supervision.

Education

Pusan National University 2018.03 ~ 2024.02
Bachelor's in Industrial Engineering

Pusan National University 2024.03 ~ 2026.02
Master's in Industrial Data Science & Engineering
Supervisor: Prof. Hyerim Bae

Research Projects

A Study on the XPL (eXplainable Process Learning) Methodology based on Artificial Intelligence 2023.03 ~ 2026.02
- *Developed an AI-based process mining methodology to create an automation tool capable of process monitoring, detection, analysis, and improvement.*
- *Main Developer*

Human Centered – Carbon Neutral Global Supply Chain Research 2023.06 ~ 2025.02
- *Developed a core technology for building an ecosystem that prioritizes safety and environmental sustainability across an integrated supply chain*
- *Sub-Developer*

Industrial Projects

Development of Process Mining and AI-based Affectionate Intelligence Technology for Customer-Specific Behavior Modeling (with LG Electronics) 2024.03 ~ present
- *Developed a methodology for managing the customer's product usage process.*
- *Main Developer*

Teaching Experience

- Tutor, **LG Electronics Process Mining Term Project** 2025.06 ~ 2025.07
- Teaching Assistant, **LGE Process Mining Classroom Training** (Supported by Celonis) 2025.06
- Teaching Assistant, **LG Electronics Process Mining Classroom Training** 2024.06
- Teaching Assistant, **Data Structures and Algorithms**, Undergraduate Course 2024.03 ~2024.07

Awards

Grand Prize 2023.08
- *PNU Industrial Artificial Intelligence Competition*

Grand Prize 2023.02
- *Graduation Project in Industrial Engineering, Pusan National University*

Conference

Only if the first author and presenter

KIIE, Korea Institute of Industrial Engineers 2025 Fall Conference 2025.11
- *Title: DHiM: A Dual-Modality Hierarchical Network for Multi-Perspective Business Process Anomaly Detection*

BPM 2025 Main Track, The 23rd International Conference on Business Process Management 2025.09
- *Title: Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies*

ICPR28, The 28th International Conference on Production Research 2025.07
- *Title: Process-Aware Prediction of Procurement Lead Time for Shipyard Delay Mitigation*

LOGMS2023, The 11th International Conference on Logistics and Maritime Systems 2023.09
- *Title: Import Container Dwell Time: Analysis of Determinant Factors with Explainable Artificial Intelligence*

Publications

Excluding Korean Publications

[1] Reasoning-Aware Proxy Reward Model using Process Mining 2026.02
***Yongjae Lee**, Taekhyun Park, Sunghyun Sim, Hyerim Bae*
arXiv preprint

[2] JustDense: Just using Dense instead of Sequence Mixer for Time Series Analysis 2025.12
*Taekhyun Park, **Yongjae Lee (co-first)**, Daesan Park, Dohee Kim, Hyerim Bae*
In Proceedings of the 2025 IEEE International Conference on Big Data (BigData)

[3] Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies 2025.08
***Yongjae Lee**, Dohee Kim, Donghwan Kim, Hyerim Bae*
Lecture Notes in Computer Science (LNCS)

[4] Identifying Key Factors influencing Import Container Dwell Time using eXplainble Artificial Intelligence 2024.12
***Yongjae Lee**, Kikun Park, Hyunjae Lee, Jongpyo Son, Seonhwan Kim, Hyerim Bae*
Maritime Transport Research (IF 3.9)

[5] Predictive Process Monitoring for Remaining Time Prediction with Transfer Learning 2024.08
*I.A. Nur, K.I. Mustafa, R.M. Hanif, Dohee Kim, **Yongjae Lee**, Hyerim Bae*
ICIC Express Letters

[6] Process-Aware Procurement Lead Time Prediction for Shipyard Delay Mitigation In Press
***Yongjae Lee**, Eunhee Park, Daesan Park, Dongho Kim, Jongho Choi, Hyerim Bae*
Submitted to selected papers of international conference on production research (ICPR)

Patents

Continual Learning Method and Device with Adaptive Memory Mechanism for Predictive Process Monitoring 2025.02
- *Hyerim Bae, I.A. Nur, **Yongjae Lee**, Dohee Kim*
- *Korean Patent, No.10-2025-0026041*

Tech Stack

Language: Python, JavaScript
Framework: PyTorch, PyTorch Geometric
Simulation: Siemens Plant Simulation
Markup: Markdown, Latex, HTML, CSS